HERMES

MERCURY

A free software HF modem and reliable data link

Email, files and messages, from below −10 dB SNR to 3.1 kbit/s

Rhizomatica • **the modem of HERMES** • **November 2026**

github.com/Rhizomatica/mercury • hermes.radio



Outline

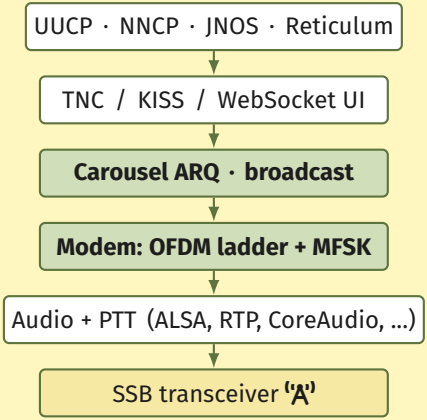
1. Mercury in HERMES
2. Modulation and coding modes
3. MFSK for weak-signal operation
4. Carousel ARQ
5. Formal methods: TLA+ and TLAPS
6. Testing and channel simulation
7. Integration and platforms
8. Outlook



MERCURY IN HERMES

Overview

- The HF software modem of **HERMES**: it turns an SSB radio into a data link for email, files and messages
- Hands the application a **TCP connection**: a VARA-compatible TNC (control 8300, data 8301), KISS and broadcast on 8100
- Or runs **in-process**: `libmercury_host` links the modem into hermes-radio-daemon
- A clean **rewrite in C** (v2, since March 2026); the C++ v1 lives on as `mercuryv1`
- **Free software**: GPL-3.0-or-later; vendored codec2 subset LGPL-2.1
- Built by Rhizomatica for communities beyond the Internet, first in the Amazon rainforest



2026 in figures

1398 commits on trunk since the C rewrite (2026-03-05)

783 commits since 1 September

8 releases since September (1.9.15 → 1.9.22)

Platforms

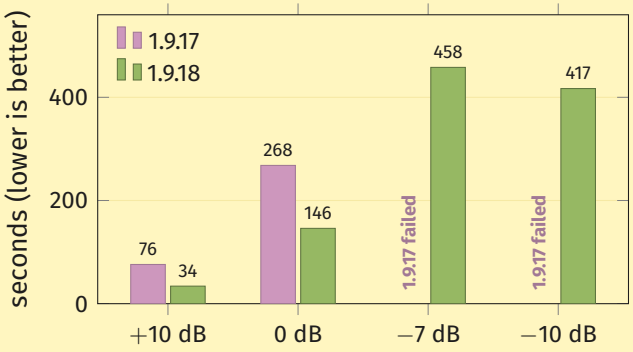
Linux (amd64, arm64, armhf, i386), Raspberry Pi 4/5, Windows (signed), macOS (universal, notarised); **iOS on the App Store**; Android and Sailfish OS as technology previews

Releases

1.9.18 (12 Oct): the carousel data plane
1.9.19 (13 Oct): MFSK centred on 1500 Hz; JNOS 2.0
1.9.20 (22 Oct): the ladder starts high and falls back fast; a 500 Hz MFSK floor; new connect frames, so no link to 1.9.19
1.9.21 (24 Oct): MFSK-only transmit; Hermes-Lite 2
1.9.22 (3 Nov): MFSK calls heard across bandwidths

Source: git history of `mercuryv2`; docs/RELEASE-1.9.18.md to docs/RELEASE-1.9.22.md

Release 1.9.18 compared with 1.9.17: a 2 KB transfer



- Two real daemons through codec2 `ch`, 2074 bytes
- **2× faster** on a good link
- Links down to −10 **dB** SNR now complete
- On air, 8 KB each way: 485/509 s → **312/238 s**

Source: docs/RELEASE-1.9.18.md



MODULATION AND CODING MODES

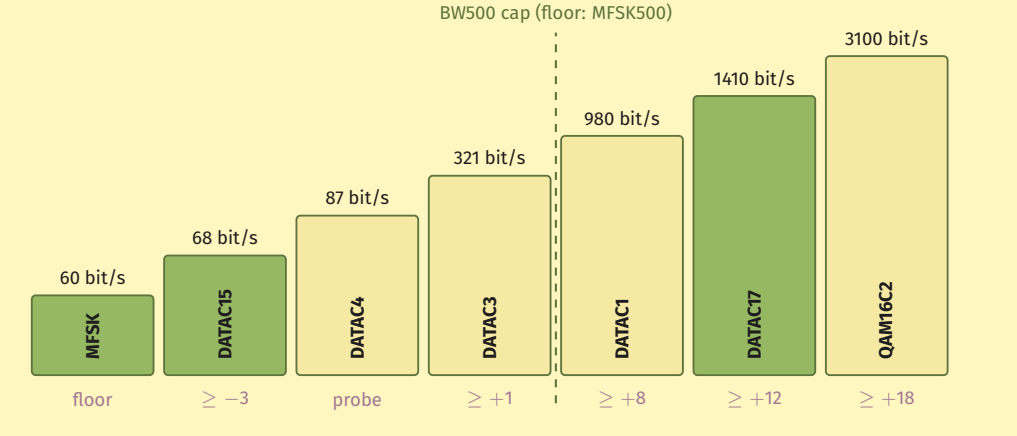
Mode overview

Mode	Origin	BW, Hz	Payload, B	bit/s	FEC (n,k)	Role
MFSK	Mercury	1000	98	∼60	LDPC (1600,800)	rung 0 (floor)
MFSK500	Mercury	500	98	∼31	LDPC (1600,800)	rung 0 at BW500
DATAC15	Mercury	200	30	68	(768,256)	rung 1
DATAC4	codec2	250	54	87	(1472,448)	rung 2
DATAC3	codec2	500	126	321	(2048,1024)	rung 3 (BW500 top)
DATAC1	codec2	1700	510	980	(8192,4096)	rung 4
DATAC17	Mercury	2100	1180	1410	(15936,9456)	rung 5
QAM16C2	codec2	2100	1213	3100	(16200,9720)	rung 6
DATAC16	Mercury	200	14	42	(640,128)	ARQ control
DATAC0/13/14	codec2	200–500	3–14	58–291	—	broadcast, tests

green = designed by us: both MFSK floors, DATAC15, DATAC17 and the control mode

Source: docs/MODES.md

Adaptive mode ladder: seven rungs from 60 to 3100 bit/s



start rung from the connect's SNR; every poll carries its SNR in dB, so a fresh reading jumps straight to its rung

Source: datalink_arq/carousel.c (car_start_level), docs/CAROUSEL-SPEC.md §15

OFDM modes designed by the project

Robust: DATAC15 and DATAC16

- Rate-1/3 LDPC (`H_256_768_22`), 200 Hz
- DATAC15: 30 bytes, 3 carriers, the first OFDM rung
- DATAC16: 14 bytes, calls, polls and ACKs
- 70/100 frames at −7 dB on MPP fading

Fast: DATAC17, on codec2's QAM16C2 code

- 2.1 kHz, about 1.2 KB a frame
- codec2's QAM16C2: 16-QAM, (16200,9720), 3100 bit/s
- DATAC17: QPSK on that codeword, 1410 bit/s
- DATAC17 97/100 at +10 dB (MPP)

Accuracy of link adaptation

Per-mode **SNR calibration** (QAM16C2 read 9 dB low); a 16-QAM estimator above 12 dB; the **TX filter** no longer cuts our outer carriers (6.5 dB); **FFT-correlation** acquisition, 2.2× cheaper

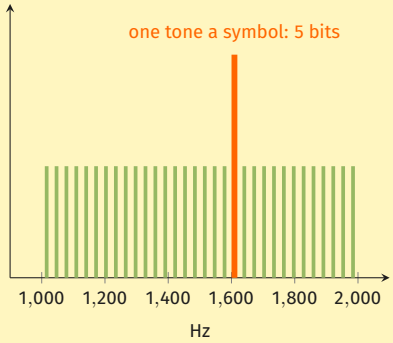
Source: docs/MODES.md, docs/OTA-PHASE-A-SNR-CALIB.md; PRs #163, #337, #350



MFSK FOR WEAK-SIGNAL OPERATION

MFSK: a non-coherent waveform below the OFDM modes

- Mercury's own modem: **32-FSK on OFDM subcarriers**, non-coherent
- 8 kHz, Nfft 256: tones 31.25 Hz apart, 40 ms symbols (25 baud)
- Centred on **1500 Hz** since 1.9.19: tones 1015.6–1984.4 Hz
- Preamble *and* postamble: a burst is found from either end
- LDPC N = 1600; five rates ported, **rate 1/2 ships** (98-byte frames, 13.5 s)
- Reducing the code rate to 1/16 gains only 2.1 dB: sensitivity is limited by synchronisation

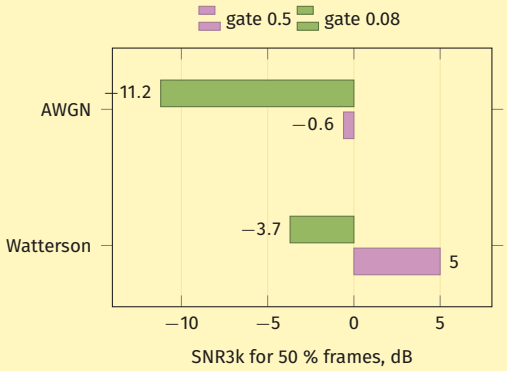


The 32 tones, 1015.6–1984.4 Hz

Source: docs/MFSK-PORT.md, modem/modem_mfsk.c, docs/RELEASE-1.9.19.md

Sensitivity limited by synchronisation, not by coding

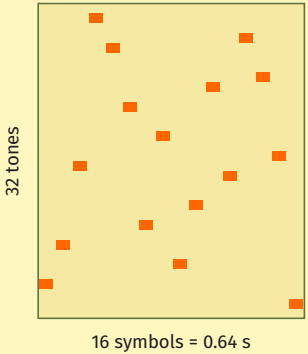
- All five code rates gave **identical** frame error rates: frames near the threshold never reached the decoder
- The sync accept gate was 0.5, which demands SNR per symbol ≥ 0 dB
- At **0.08**, with false anchors released:
  - AWGN 50 % point: $-0.6 \rightarrow -11.2$ **dB**
  - Watterson 2 ms / 1 Hz: $+5.0 \rightarrow -3.7$ **dB**
- Dual-ended acquisition contributes about 2 dB; codeword interleaving adds 19 % throughput at the margin



Source: docs/MFSK-PORT.md

Pattern acknowledgements: 0.64 s of airtime

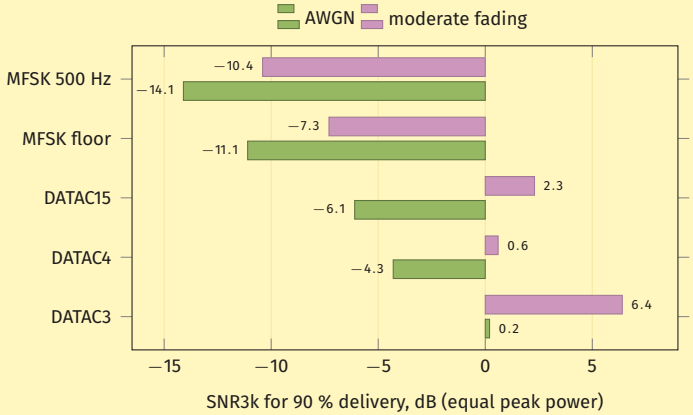
- An acknowledgement as a **Welch-Costas pattern**: 16 MFSK symbols, **0.64 s**
- 4.8× less airtime than a 3.08 s DATAC16 frame
- Detected every time down to about −9 dB on fading and −22 dB on AWGN
- **0 false alarms** in 30 trials down to −26 dB; 0 in 2 h of noise after the half-subcarrier search
- Frequency search over ±78 Hz: an IC-7100 and an sBitx went from 0/5 to 17/17 patterns on air



(illustrative layout: one tone per symbol, no repeats)

Source: docs/MFSK-PORT.md, README.md; PRs #160, #299

Sensitivity of MFSK compared with the OFDM modes



- MFSK operates **5 to 8 dB** below DATAC15 on AWGN, **10 to 13 dB** under moderate fading
- Against modem73's coherent RFDM modes: 2.7 dB better per bit on AWGN, a tie under fading

Sim, 100 trials a point, Watterson model, 2026-10-15 (PR #390); docs/MFSK-PORT.md; docs/RELEASE-1.9.19.md

A 500 Hz MFSK floor for BW500 sessions

MFSK500: the BW500 floor (1.9.20)

- The 1 kHz MFSK cannot be decoded behind a 500 Hz filter
- 32 tones 15.625 Hz apart, 80 ms symbols, 26 s frames
- 90 %: −14.1 **dB** AWGN, −10.4 dB moderate fading, behind the filter too
- Searches at 2 kHz: idle receive **cheaper** than the 1 kHz floor
- BW2300 and BW2750 keep the 1 kHz floor

Receiver fixes, all MFSK modes

- Dial offset to an eighth of a subcarrier
- Pattern search in half-subcarrier steps (removes a 3–4 dB loss)
- LLR noise from the tones' own bins
- A burst with a **gap in its audio** (a playback underrun at the transmitter) is read across the gap

Calls across bandwidths (1.9.22)

A waiting station decodes **both** floors, for 1.3–2.3 % of a Pi 4 core. On air, BW2750 against BW500 with an MFSK-only caller: **0 of 4** connects before, **5 of 5** after

Source: docs/RELEASE-1.9.20.md, docs/RELEASE-1.9.22.md; PRs #406, #451



CAROUSEL ARQ

Motivation for a new ARQ

Previous design: stop-and-wait

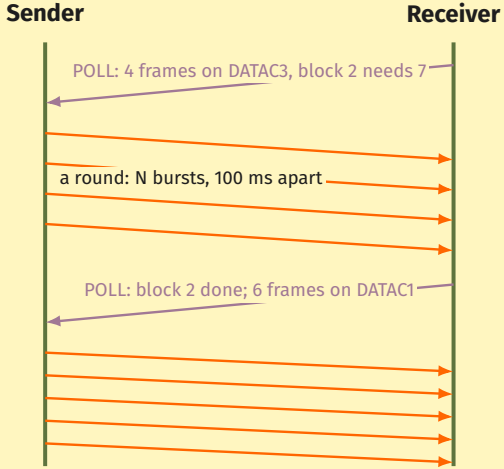
- SNR-driven: one frame, one ACK, one turnaround
- Rungs chosen from SNR estimates that were wrong per mode
- Lost ACKs on the weak reverse path stalled the session
- Sim NVIS: **0 of 20** runs completed

Carousel design

- **Receiver-driven**: the receiver requests what it still needs
- **Erasure-coded**: any k of the pieces rebuild a block, so no retransmission of a particular frame
- No per-frame ACKs, no sequence numbers
- Rungs chosen by **measured goodput**
- Merged 3 Oct (#322), the only data plane since 11 Oct (#364)

Source: docs/CAROUSEL-ARQ.md, docs/CAROUSEL-REPORT.md

Carousel operation



- Systematic Cauchy **Reed-Solomon** over GF(256), blocks of up to 96 pieces
- **23–24 byte pieces**: four fit one MFSK frame
- Up to 8 blocks open; each ends in a seed-keyed CRC-32 and fails closed
- Four consecutive losses mark a rung unusable; it is probed again after 60 s, the interval doubling up to 8 min

Source: docs/CAROUSEL-SPEC.md, docs/CAROUSEL-ARQ.md

Channel access, connection set-up and versioning

Single timing master

The caller owns every timer; the callee keys only in its slots. **0 collisions** in 960 two-way deep-fade sim runs (#351)

Connection at low SNR

Fast DATAC16 CALLs first (2–4, from `RETRIES`), then MFSK. A callee hearing a CALL twice answers on MFSK. The ACCEPT names the start rung and carries the callee's nonce

Session binding

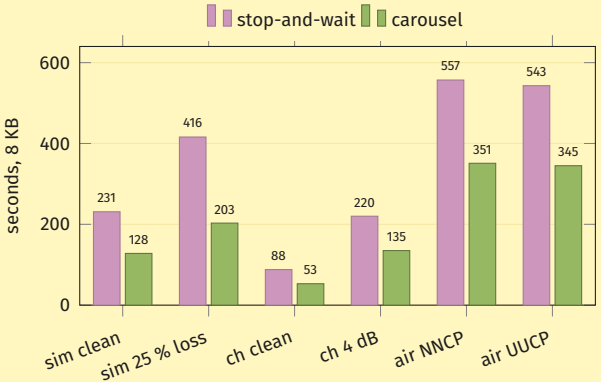
Seed = CRC16(session id, nonce, both calls), mixed into every frame's CRC. Since 1.9.20 the connect frames have their own type (CALL2): 1.9.20–1.9.22 interoperate; a 1.9.19 CALL is refused

Listen before talk

Every keying path waits for a clear channel; the busy detector is opt-in (off by default)

Source: docs/CAROUSEL-TURNS.md, docs/CAROUSEL-SPEC.md, docs/RELEASE-1.9.20.md; PRs #317, #351, #420, #443

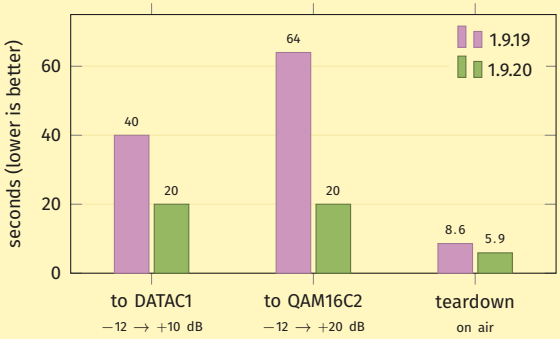
Results: simulation, modem bench and on-air tests



- **35–45 % shorter** transfer times in every measurement
- Simulated NVIS: stop-and-wait never completed; the carousel did
- On air: mean of two runs in each direction, sBitx test bench

Source: docs/CAROUSEL-REPORT.md, docs/CAROUSEL-ARQ.md

Release 1.9.20: start high, fall back fast



- Every poll carries its **SNR in dB**: both ends know both directions, and a reading jumps straight to its rung
- A drop below the control mode reaches the floor in **96 s**, not 15 min; 120/120 simulated 50 KB transfers, against 92
- **High Doppler** (10–30 Hz, no OFDM mode decodes): starts on the floor; ITU-R/CCIR presets **60/84 → 80/84**
- Idle sessions with a fifth of the frames lost: 5 of 100 dropped in 30 min, against 82

Simulator, 120 seeds a cell; Go harness through the Watterson presets; on air, sBitx bench. Source: docs/RELEASE-1.9.20.md; PRs #406, #413



FORMAL METHODS: TLA+ AND TLAPS

Formal models of the protocol layers

Model	What it proves	Checked
CarouselBlocks	InOrder, Integrity, RetireSafe, NoPhantom: delivery in order, nothing invented, nothing retired early	TLC up to **63.4 M states**; an adversary config at 12.1 M
Liveness	Completes: every block is delivered, under a lossy adversary	2.58 M states, 14.2 h
CarouselProofs	Spec $\Rightarrow \Box$Safety for any number of blocks	**TLAPS: 823 obligations** (Z3, Zenon)
CarouselTurns (timed)	NoOverlap: the two stations never key together; MNotShutOut	18.5 M states; MFSK config 27.1 M
RungAgreement	Agree, WastedF, FallbacksAgree, Delivered: both ends agree on the rung	3234 states; 17168 at 4 rungs
ConnectHandshake	NoDoom, NoHalfConnect, Settles: no connect the other end does not stand by	9473 states; 34177 big
SessionTeardown	NoKill, NoLostWord, NoCut, Mutual: no end leaves on another session's frame or cuts the last word	2.57 M states lossy; 15.6 M with foreign patterns

Source: specs/carousel, specs/turns, specs/rungs, specs/connect, specs/teardown (README files)

Validation of the models

Mutation testing

- Turns: five mutants (no grid, late, long, blind, reuse) each violate NoOverlap
- Rungs: five more (no request, no bind, deaf, unnamed, previous rung), and the code before #448
- Blocks: the no-check config violates Integrity, as it must
- Before the integrity work, the model found **two real bugs**, 9 steps each

Conformance of the implementation

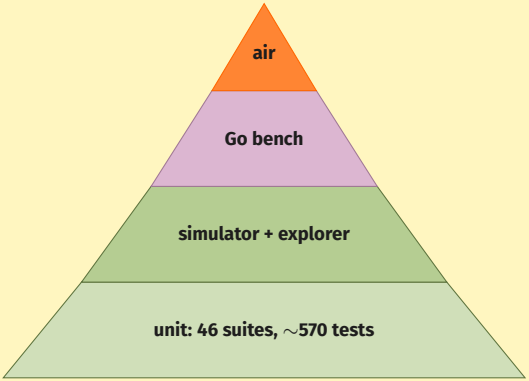
- `car_explore` runs the **real** `carousel.c` under every set of up to 3 losses or sensing failures, plus spurious BREAKs and corrupt frames
- **58 179 runs in about 10 s**, on every CI run; 9.09 M in the long sweep
- Overrun and slot oracles in the simulator check the timed model's assumptions

Source: specs/, tests/sim/car_explore.c; PRs #345–#347



TESTING AND CHANNEL SIMULATION

Test framework



- **Unit**: modems, LDPC, carousel, ARQ, TNC, JSON, rings, channel model
- **Simulator**: two state machines on a virtual clock; A/B benches on 120 seeds
- **Go bench**: two real daemons over FIFO, socket or RTP audio through `ch` or Watterson; two different builds can be paired
- **On air**: sBitx, IC-7100 and FT-710 stations, UUCP / NNCP / JNOS transfers
- **One framework**: no second copy of test data, no tool CI cannot run

Source: tests/Makefile, tests/integration, tests/sim

Continuous integration on every pull request

Builds

Linux amd64, arm64, armhf; Windows x64 (cross); macOS arm64; Android; the Fyne GUI on Linux (amd64, arm64) and Windows

Sanitizers

ThreadSanitizer on the unit tests; AddressSanitizer + UBSan on the unit tests and a full integration session

Tests

Unit tests on amd64 and **i386**; the Go integration bench, also on a fresh random channel seed each run, every seed logged

Additional tooling

libFuzzer targets for the ARQ header, frame header, KISS, arithmetic and WebSocket JSON parsers; signed and notarised release pipeline

Source: .github/workflows, docs/SANITIZERS.md, docs/FUZZING.md

Channel model compliant with ITU-R F.1487

- Watterson model with the **Gaussian Doppler spectrum** of **ITU-R F.1487** / **CCIR 520-2**: spread = 2σ
- Named presets `ccir-good/moderate/poor/flutter` and ten `itur-*` (low to high latitude, NVIS)
- An exact delay line; the former Butterworth presets are retained for reproducibility
- The former “poor” preset faded **twice as fast** as CCIR Poor: 1.007 against 0.515 fades per second

Acceptance tests in the CI suite

	Measured	Theory
Power beyond 3σ	0.273 %	0.27 %
Power beyond the spread	4.68 %	4.55 %
Fades below -10 dB, s^{-1}	0.5067	0.5072

Reference values: the Gaussian spectrum and Rice’s closed-form level-crossing rate. Records of 900 s with fixed seeds.

Cross-validation with MATLAB

Every preset compared with the R2026a Communications Toolbox (`stdchan`, `comm.RayleighChannel`): **26 of 26 checks passed**; a second run reproduced the results byte for byte.

Source: PR #379 (Joseph Freivald, merged 19 Oct): docs/CHANNEL-MODEL.md, tests/common/test_watterson.c

Independent measurement tools

Channel simulators

- codec2 `ch`: AWGN and MPP
- Our Watterson model (above)
- AE4JY's **PathSim**, calibrated against ours to $\pm$0.2 dB
- OpenARQ's **skywave** harness, with a Mercury adapter

Probes and sweeps

- Pattern, acquisition and burst-train probes
- MFSK channel sweeps by file, through any channel tool
- An opt-in RX audio dump: a frame missed on air can be decoded offline
- On-air recordings with noise added offline: the bench's own fringe curve

Source: utils/, docs/MFSK-PORT.md; PR #389



INTEGRATION AND PLATFORMS

Supported applications

Mail and files

UUCP is the HERMES default; NNCP works too; JNOS 2.0 BBS-to-BBS; Winlink through Pat and scanning BPQ32 gateways; vARIM (`max_keydown_s`)

Networks

Reticulum over KISS on the broadcast port; broadcast files by a RaptorQ fountain, wire-compatible with hermes-broadcast; HERMES station beacons (in review, #480)

Encryption (in review)

Optional Noise_KK sessions (X25519, ChaCha20-Poly1305), XChaCha20 broadcast files; off by default (PR #410)

Source: docs/JNOS.md, docs/RETICULUM.md, docs/BROADCAST-FILE.md, docs/TNC.md; PRs #410, #480

Interfaces and platforms

- **Our own WebSocket server**: 28 820 vendored GPL-2.0-only lines replaced by about 1500 of ours (RFC 6455, TLS through OpenSSL)
- Status: modes, the peer's view of our signal, frequency, **carrier offset**
- **TUNE**: a 1 kHz tuning carrier, hard 60 s unkey deadline
- Audio: ALSA, Pulse, CoreAudio, AAudio, WASAPI, DirectSound, **RTP** to hermes-radio-daemon, **Hermes-Lite 2**
- PTT: Hamlib (also driverless USB to CP210x radios), serial RTS, CM108 GPIO, HERMES shared memory
- **Fyne GUI** with the engine linked in
- **Windows**: Authenticode-signed (Certum)
- **macOS**: universal x86_64 + arm64, static Hamlib, notarised; a Mac App Store package
- **iOS**: on the App Store; **Android** (.apk) and **Sailfish OS**: technology previews
- **Raspberry Pi** 4 and 5, the HERMES station computers

Source: README.md, docs/WEBSOCKET_STATUS.md, docs/HERMES-LITE-2.md, docs/RELEASE-1.9.20.md, docs/RELEASE-1.9.21.md; PRs #383, #418, #421, #428



OUTLOOK

Future work

Optional encryption

Encrypted sessions on the carousel plane, off by default (Noise_KK, PR #410)

Station beacons

HERMES position and status beacons on DATAC16, heard by every listening station (PR #480, received 6 of 6 on air)

Mobile apps

Android and Sailfish OS out of preview; a background service on Android

HERMES

THANK YOU

- github.com/Rhizomatica/mercury
- hermes.radio
- docs/ in the repository: modes, carousel, specs, channel model

Mercury is free software (GPL-3.0-or-later).

This presentation: CC BY-SA 4.0.

